

Finite Paths Give a Counterexample to the Componentwise Cardinality Criterion

Automatic mathematics run `fa_banach_001`

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Status. Candidate counterexample, likely valid, for human review. It gives a full negative answer to the finite-connected-component question immediately after Theorem 1.2 of Baudier–Braga–Farah–Vignati–Willett.

1 Source question

The source is F. P. Baudier, B. M. Braga, I. Farah, A. Vignati, and R. Willett, *Coarse equivalence versus bijective coarse equivalence of expander graphs*, arXiv:2307.11529, published in *Mathematische Zeitschrift* 307 (2024), Paper 44 [1].

Their Theorem 1.2 concerns coarse disjoint unions $X = \bigsqcup_n X_n$ and $Y = \bigsqcup_n Y_n$ of expander graphs and a coarse equivalence $f : X \rightarrow Y$. It says that the following are equivalent:

1. f is close to a bijective coarse equivalence;
2. after discarding finitely many components, f matches components of equal cardinality, with the total cardinalities of the discarded parts also equal.

Immediately afterward the authors write:

“We do not know whether the conclusion of Theorem 1.2 can fail if X and Y are coarse disjoint unions of finite connected graphs (that are not necessarily expanders).”

This occurs across printed pages 3–4. The complete source evidence follows.

Printed page 3 (end):

try of the underlying sequence. We should note that a coarse disjoint union of finite graphs is always amenable, so Theorem 1.1 does not apply to such spaces.

Our main result is as follows. It should be thought of as an analog of Theorem 1.1 showing that the conclusion holds under an obvious necessary condition.

Theorem 1.2. *Let X and Y be coarse disjoint unions of expander graphs, say $X = \bigsqcup_n X_n$ and $Y = \bigsqcup_n Y_n$. Let $f: X \rightarrow Y$ be a coarse equivalence. The following are equivalent.*

1. *The map f is close to a bijective coarse equivalence.*
2. *There are cofinite subsets N, M of $\mathbb{N}$ and a bijection $i: N \rightarrow M$ such that*

$$\sum_{n \in \mathbb{N} \setminus N} |X_n| = \sum_{n \in \mathbb{N} \setminus M} |Y_n|$$

and for all $n \in N$

$$|X_n| = |Y_{i(n)}| \quad \text{and} \quad f(X_n) \subseteq Y_{i(n)}.$$

We do not know whether the conclusion of Theorem 1.2 can fail if X and Y are coarse disjoint unions of finite connected graphs (that are not

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4

BAUDIER, BRAGA, FARAH, VIGNATI, AND WILLETT

necessarily expanders). The statement of Theorem 1.2 is purely combinatorial/geometric, but the proof requires uniformly finite homology [BW92] as exploited by Whyte in [Why99].

2 Definitions

Two maps $u, v: S \rightarrow T$ into a metric space are *close* if $\sup_{x \in S} d(u(x), v(x)) < \infty$. A map $u: S \rightarrow T$ is *coarse* if for every $r > 0$ there is $s > 0$ such that

$$d_S(x, x') < r \implies d_T(u(x), u(x')) < s.$$

It is a *coarse equivalence* if it has a coarse inverse up to closeness. A *bijective coarse equivalence* is a coarse equivalence that is also a bijection.

A coarse disjoint union of finite metric spaces (P_n) is their disjoint union with a metric restricting to the given metric on each component and satisfying $d(P_n, P_m) \rightarrow \infty$ as $(n, m) \rightarrow \infty$ with $n \neq m$.

3 Idea of the proof

Equal component cardinalities control only total mass; expansion is what lets the source theorem redistribute that mass by a uniformly bounded matching. Paths have large pieces with tiny boundary, so one can create a persistent local mass deficit while compensating elsewhere in the same component.

On a path of $4n$ vertices, compress the first half by a factor of two. Map the second half with slope $3/2$, so that the total image remains uniformly dense. The resulting maps and explicit inverse sections are uniformly Lipschitz and inverse to within one, hence define a coarse self-equivalence of the disjoint union. Component sizes match exactly. But any map within a fixed distance C must send the first $2n$ vertices into only $n + O(C)$ possible targets. No injection, and therefore no bijection, can do this for arbitrarily large n .

4 Construction

For $n \geq 1$, let

$$P_n = \{0, 1, \dots, 4n - 1\}$$

with the path metric. Put $X = \bigsqcup_{n \geq 1} P_n$ and define

$$d((n, k), (m, j)) = \begin{cases} |k - j|, & n = m, \\ 4n + 4m, & n \neq m. \end{cases} \quad (1)$$

For $k \in P_n$, define

$$f_n(k) = \begin{cases} \lfloor k/2 \rfloor, & 0 \leq k < 2n, \\ n + \lfloor 3(k - 2n)/2 \rfloor, & 2n \leq k < 4n, \end{cases} \quad (2)$$

and, for $j \in P_n$, define

$$g_n(j) = \begin{cases} 2j, & 0 \leq j < n, \\ 2n + \lfloor 2(j - n)/3 \rfloor, & n \leq j < 4n. \end{cases} \quad (3)$$

Let $f, g : X \rightarrow X$ act by f_n, g_n on P_n .

Lemma 1. *Equation (1) defines a uniformly locally finite metric, and X is a coarse disjoint union of the finite connected graphs P_n .*

Proof. The only nontrivial triangle check is when two vertices lie in one component and the third lies in another. Since $\text{diam}(P_n) = 4n - 1$ while the two cross-component legs have sum at least $8n + 8$, the triangle inequality holds. The restriction to P_n is its path metric, and $d(P_n, P_m) = 4n + 4m \rightarrow \infty$ off the diagonal, so this is a coarse disjoint union.

Fix $R > 0$. A ball of radius R meets at most $2\lfloor R \rfloor + 1$ vertices in its own path. If it meets another P_m , then $4n + 4m \leq R$, hence $m \leq R/4$. Thus its cardinality is bounded by

$$2\lfloor R \rfloor + 1 + \sum_{m \leq R/4} 4m,$$

independently of its center. Hence X is uniformly locally finite. $\square$

Proposition 2. *The maps f and g are 2-Lipschitz and*

$$d(g(f(x)), x) \leq 1, \quad d(f(g(x)), x) \leq 1 \quad (x \in X).$$

Consequently f is a coarse equivalence with coarse inverse g .

Proof. Along a component, consecutive values of f_n differ by at most 2: the first part of (2) has increments 0 or 1, the transition has increment 1, and the second part has increments 1 or 2. Consecutive values of g_n differ by at most 2 as well. Between distinct components, both maps preserve the component indices, so they preserve the distance $4n + 4m$ exactly. Thus both maps are globally 2-Lipschitz.

For $k < 2n$,

$$g_n(f_n(k)) = 2\lfloor k/2 \rfloor,$$

which differs from k by at most 1. If $k = 2n + r$, then

$$g_n(f_n(k)) = 2n + \left\lfloor \frac{2\lfloor 3r/2 \rfloor}{3} \right\rfloor,$$

whose final term equals r for even r and $r - 1$ for odd r .

Similarly, $f_n(g_n(j)) = j$ for $j < n$. If $j = n + s$, then

$$f_n(g_n(j)) = n + \left\lfloor \frac{3\lfloor 2s/3 \rfloor}{2} \right\rfloor.$$

Writing s modulo 3 shows that the final term is s when $s \equiv 0 \pmod{3}$ and $s - 1$ otherwise. Both composition errors are therefore at most 1. $\square$

5 The bounded-displacement obstruction

Theorem 3. *The coarse equivalence $f : X \rightarrow X$ satisfies condition (2) of the source Theorem 1.2 with all components retained and the identity matching of components. Nevertheless, f is not close to any bijection $X \rightarrow X$. In particular, it is not close to any bijective coarse equivalence.*

Proof. The component condition is immediate: $f(P_n) \subseteq P_n$ and $|P_n| = |P_n|$ for every n . In the notation of the source theorem, take $N = M = \mathbb{N}$ and $i(n) = n$; both discarded sums are zero.

Suppose, toward a contradiction, that a bijection $h : X \rightarrow X$ and a constant $C < \infty$ satisfy

$$d(h(x), f(x)) \leq C \quad (x \in X). \tag{4}$$

Choose an integer $n > \max\{C, 1\}$. Since $f(P_n) \subseteq P_n$ and every point outside P_n is at distance at least $4n + 4 > C$ from P_n , (4) implies $h(P_n) \subseteq P_n$.

Let

$$A_n = \{0, \dots, 2n - 1\} \subseteq P_n, \quad B_n = \{0, \dots, n - 1\} \subseteq P_n.$$

By (2), $f(A_n) = B_n$. Equation (4) therefore forces

$$h(A_n) \subseteq N_C(B_n) \cap P_n \subseteq \{0, \dots, n - 1 + \lfloor C \rfloor\}.$$

The set on the right has at most $n + \lfloor C \rfloor < 2n = |A_n|$ elements. This contradicts injectivity of h . $\square$

Corollary 4 (Negative answer to the source question). *The conclusion of the source Theorem 1.2 fails for coarse disjoint unions of finite connected graphs. It already fails when every component is a path and $X = Y$.*

6 Verification notes

The proof uses only the displayed formulas. The accompanying script `code/check_finite_path_counterexample.py` checks every component $1 \leq n \leq 2000$. It verifies that both adjacent-step bounds equal 2, both composition errors are at most 1, and every component image is 1-dense. It also checks the final cardinality deficit for integer closeness bounds $0 \leq C \leq 500$. These computations are sanity checks, not part of the proof.

7 Limitations and bounded novelty audit

This packet answers only the combinatorial/geometric question after Theorem 1.2. It does not claim that the counterexample is induced by an isomorphism of uniform Roe algebras. The latter rigidity conjecture, stated separately as Conjecture 1.3 in the source, has independent later resolutions and is not the target here.

On 9 August 2026, the four run indexes were searched for arXiv:2307.11529, the exact question phrase, and the core terms “coarse disjoint union”, “finite connected graphs”, and “bijective coarse equivalence”. Exact and keyword web/arXiv searches were also run. The latest source page still contains the question. Finally, the OpenAlex record, updated 23 July 2026, listed two citing works; inspection of their titles, abstracts, and searchable text found only operator-algebraic applications to expanders, not the present finite-path construction or an answer to this question. No prior answer was found within these bounds. This supports plausible novelty but is not a certification.

8 Human-review recommendation

Recommendation: promote after expert confirmation. The main review points are: (i) that the source’s phrase “the conclusion of Theorem 1.2 can fail” is correctly read as asking whether the stated equivalence can fail; (ii) the global metric in (1) meets the source definition of coarse disjoint union; and (iii) the counting argument excludes every bounded-distance bijection, not merely componentwise permutations.

References

- [1] F. P. Baudier, B. M. Braga, I. Farah, A. Vignati, and R. Willett, *Coarse equivalence versus bijective coarse equivalence of expander graphs*, *Mathematische Zeitschrift* 307 (2024), Paper 44; arXiv:2307.11529.